HORIBA
Semiconductor

Back side wafer cooling system

GR-300 Series

Pressure control with high stability and accuracy

A high precision piezo valve is used to control pressure even with very low differential pressure. A highly accurate pressure sensor is installed to measure the outlet pressure within <1% F.S.

Using mass flow sensor (Option)

By applying the well established fluid control technology from HORIBA STEC's mass flow products, the product enables minute differential pressure control with the addition of gas flow monitoring (with mass flow sensor) for self diagnostic capability.

RoHS compliance

The GR-300 Series uses parts that do not damage the environment and comply with the latest RoHS regulations.

System configuration example

When using GR-300 for back side wafer cooling, accurate pressure control of helium gas for both the wafer center and wafer edge is possible. Moreover, the gas flow can be monitored accurately with the integrated mass flow sensor.

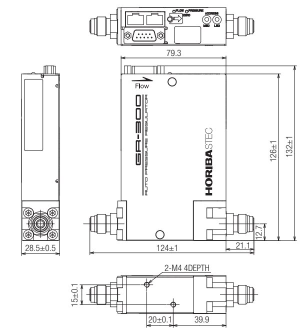
■ Product specifications

Model	GR-312	GR-312F	GR-314	GR-314F	GR-317	GR-317F
Valve Type	C : Nomally close					
Pressure control range	1-100% F.S.					
Pressure accuracy	0.5% F.S.					
Operating temperature	5-50℃ (Recommended temperature : 15-40℃)					
Response	≤ 1 second (in an adjustment condition)					
The maximum operating pressure	250 kPa(A)					
Pressure resistance	300 kPa(A)					
Leak integrity	≤ 5×10^{-12} Pa·m³/s(He)					
Pressure rate output signal	0-10VDC		0-100%			
Power supply	+15V±5% 150mA -15V±5% 150mA		Applicable for ODVA standard, DC24V 4.0VA		24VDC±4V 6.6VA	
Digital Interface	RS-485 F-Net Protocol		DeviceNet™ Protocol		EtherCAT® Protocol	
Gas *1	–	He, Ar, N_2	–	He, Ar, N_2	–	He, Ar, N_2
Presure full scale	13.33kPa (A) (100Torr)					
Flow full scale	–	20, 50, 100 SCCM	–	20, 50, 100 SCCM	–	20, 50, 100 SCCM
Flow rate measurement range	–	0-100% F.S.	–	0-100% F.S.	–	0-100% F.S.
Flow accuracy *2	–	±1.0 %R.S. (Flow rate>25% F.S.) ±0.25 %F.S. (Flow rate≤ 25% F.S.)	–	±1.0 %R.S. (Flow rate>25% F.S.) ±0.25 %F.S. (Flow rate≤ 25% F.S.)	–	±1.0 %R.S. (Flow rate>25% F.S.) ±0.25 %F.S. (Flow rate≤ 25% F.S.)
Flow rate output signal	–	0-5VDC	–	0-100%	–	0-100%
Standard fittings	1/4 inch VCR equivalent					
Mounting orientation	Free					

*1 Please consult about other specifications.
*2 Flow accuracy is guaranteed only for calibration gas and flow rate of full scale.
Temperature range in which "accuracy" is guaranteed is in accordance with SEMI.
SCCM denote gas flow rate in ml/min, respectively.
F.S. : Full scale, R.S. : Reading scale, (A) : Absolute Pressure

■ External dimensions

GR-312 / 312F



Please consult about other models.

HORIBA STEC

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http://www.horiba.com/horiba-stec/

Please read the operation manual before using this product to ensure safe and proper handling of the product.

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